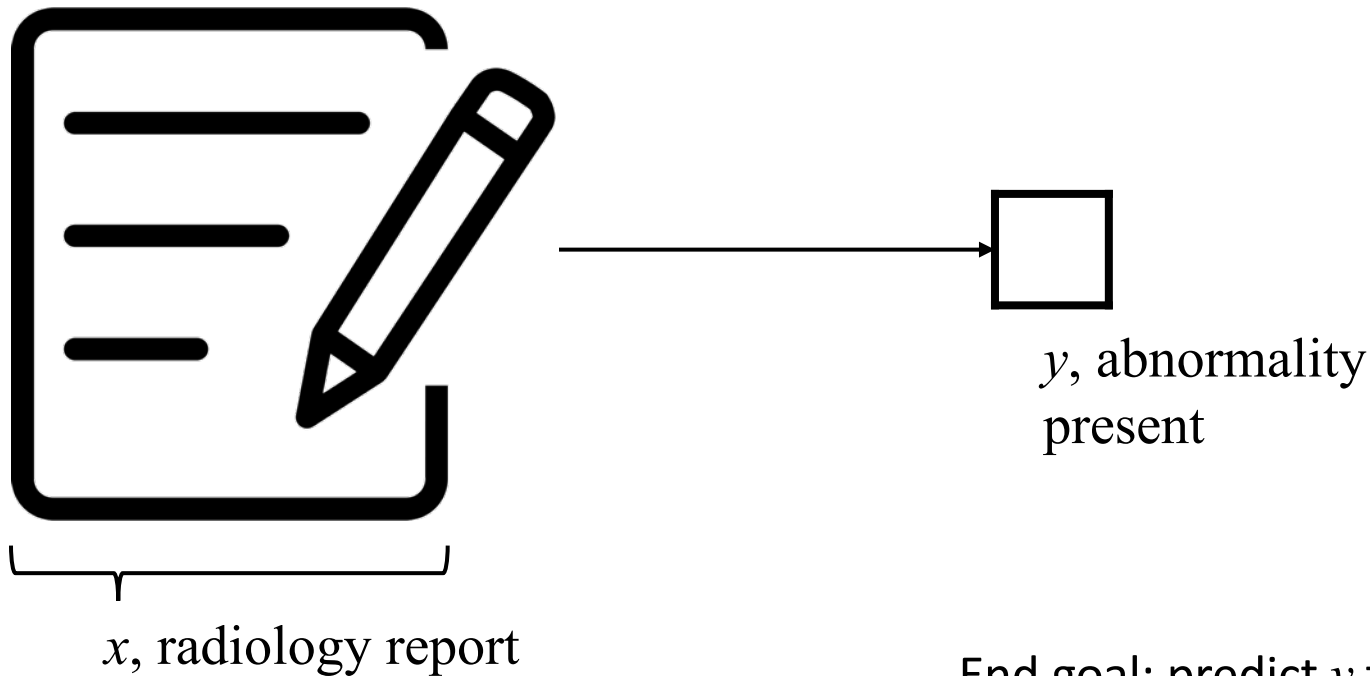


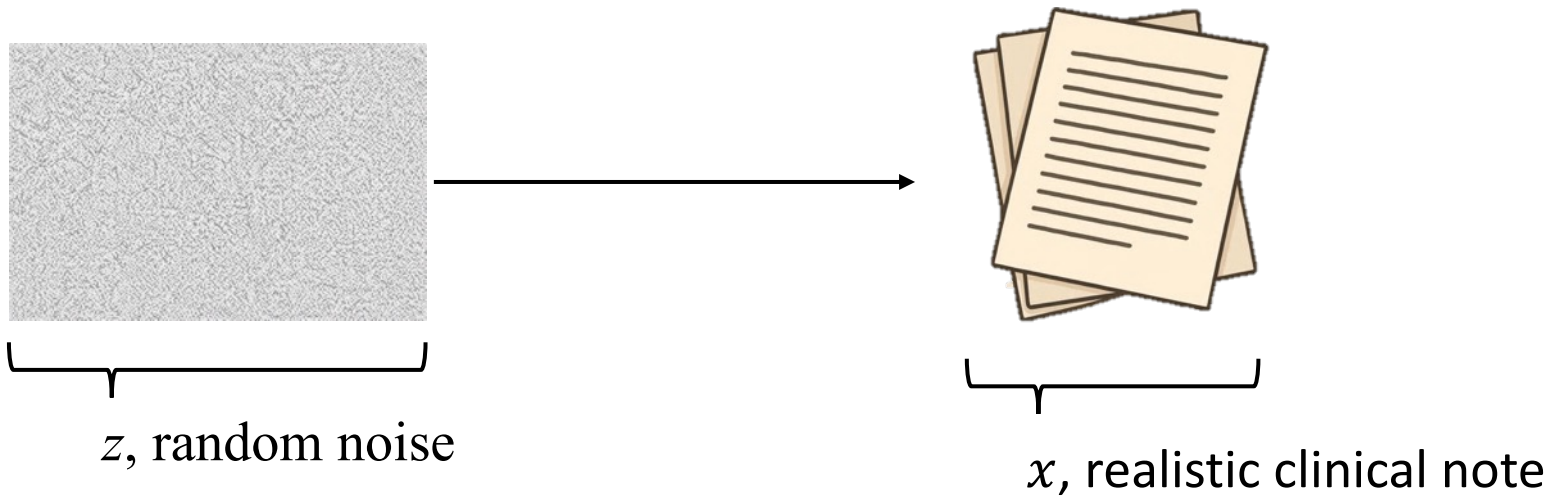
From Bags of Words to Large Language Models

Matthew Engelhard

Problem 1: make predictions from text

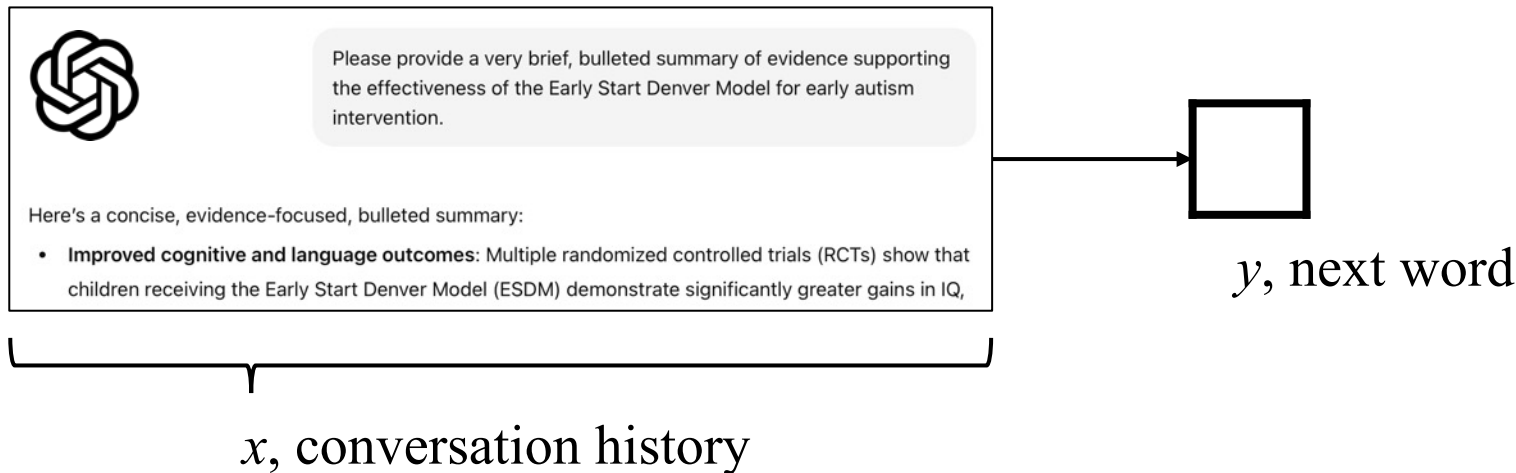


Problem 2: generate text



End goal: create structure from randomness

Is this generative or predictive?



“next-token prediction”

The central problem of natural
language processing:

*How do we convert text to numbers
our models can work with?*

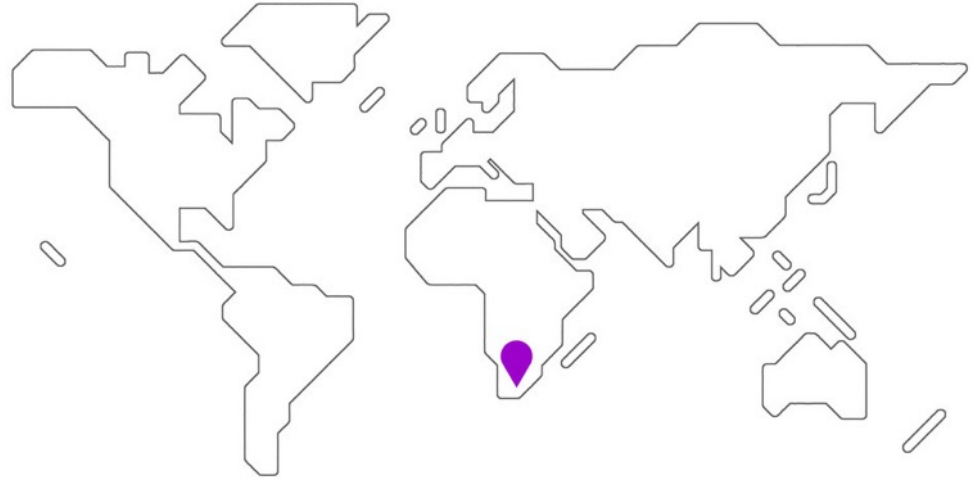
Text to numbers, v1: word counts

A count-based model for maternal health triage

Case Study: SMS Triage for Global Maternal Health

Maternal Health HelpDesk:

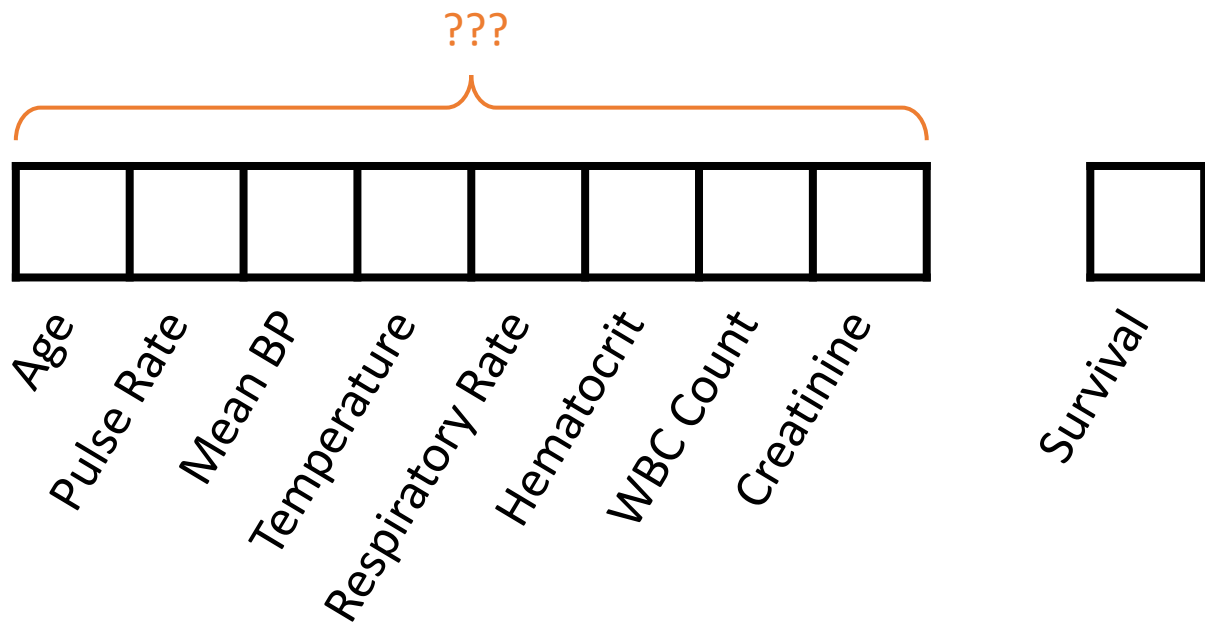
**2 million women connected to
NDoH staff via SMS**



<https://www.praekelt.org>

Binary Classification: Urgent Message? (Yes/No)

A Simple Predictive Model: ICU Mortality



End goal: predict odds of hospital mortality

Training Set (Historical Data)

x_1

x_2

x_3

x_4

$\vdots$

x_{N-1}

x_N

y_1

y_2

y_3

y_4

$\vdots$

y_{N-1}

y_N

Find an equation that
predicts y based on x
across the training set

Making Predictions for New x

 x_1 y_1 x_2 y_2 x_3 y_3 x_4 y_4 $\vdots$ $\vdots$ x_{N-1} y_{N-1} x_N y_N

Find an equation that
predicts y based on x
across the training set

 x_{N+1} y_{N+1}

<- Learn to predict new y

Case Study: SMS Triage for Global Maternal Health



<https://www.praekelt.org>

Can we use a standard predictive model
setup to solve this problem?

This time, our training data is text

x_1	What helps with morning sickness?	y_1	y_i : Urgent or Not Urgent?
x_2	How many months should I breastfeed?	y_2	
x_3	I passed out and Mom said I was shaking	y_3	
x_4	Where is the nearest clinic?	y_4	
	$\vdots$	$\vdots$	
x_{N-1}	I am having heavy bleeding, what should I do?	y_{N-1}	
x_N	What foods should I eat while pregnant?	y_N	

x_{N+1}	My heart is racing and I can't catch my breath	y_{N+1}	<- Learn to predict new y
-----------	--	-----------	-----------------------------

We need numbers, not words

- **Can we convert our text to a vector or sequence of numbers?**
- If yes, we can use logistic regression (or any other predictive model)!

First try: count words in each SMS

Step 1: Define a vocabulary of words

x_1

What helps with morning sickness?

x_2

How many months should I breastfeed?

x_3

I passed out and Mom said I was shaking

x_4

Where is the nearest clinic?

list of all words
(in no particular order)

shaking
what
clinic
how
helps
was
nearest
many

with
said
months
the
morning
mom
should
sickness

and
I
is
how
out
breastfeed
passed
where

Step 2: count how many times each vocabulary word appears in a given SMS

What helps with morning sickness?

x_I

0	1	0	0	1	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0
shaking	what	clinic	how	helps	was	nearest	many	with	said	months	the	morning	mom	should	sickness	and	I	is	how	out	breastfeed	passed	where

Step 2: count how many times each vocabulary word appears in a given SMS

I passed out and Mom said I was shaking

x_3

1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	2	0	0	1	0	1	0
shaking	what	clinic	how	helps	was	nearest	many	with	said	months	the	morning	mom	should	sickness	and	I	is	how	out	breastfeed	passed	where

Note that word order does not matter!

clinic is where nearest the

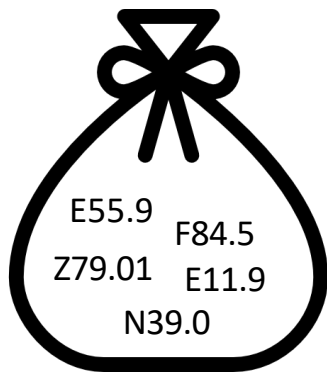
x_4

0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	1
shaking	what	clinic	how	helps	was	nearest	many	with	said	months	the	morning	mom	should	sickness	and	I	is	how	out	breastfeed	passed	where

A “bag of words”



A “bag of medical codes”



(you did this in coding assignment 1)

PCOR_RSH_DIAGNOSIS_VW EXAMPLE

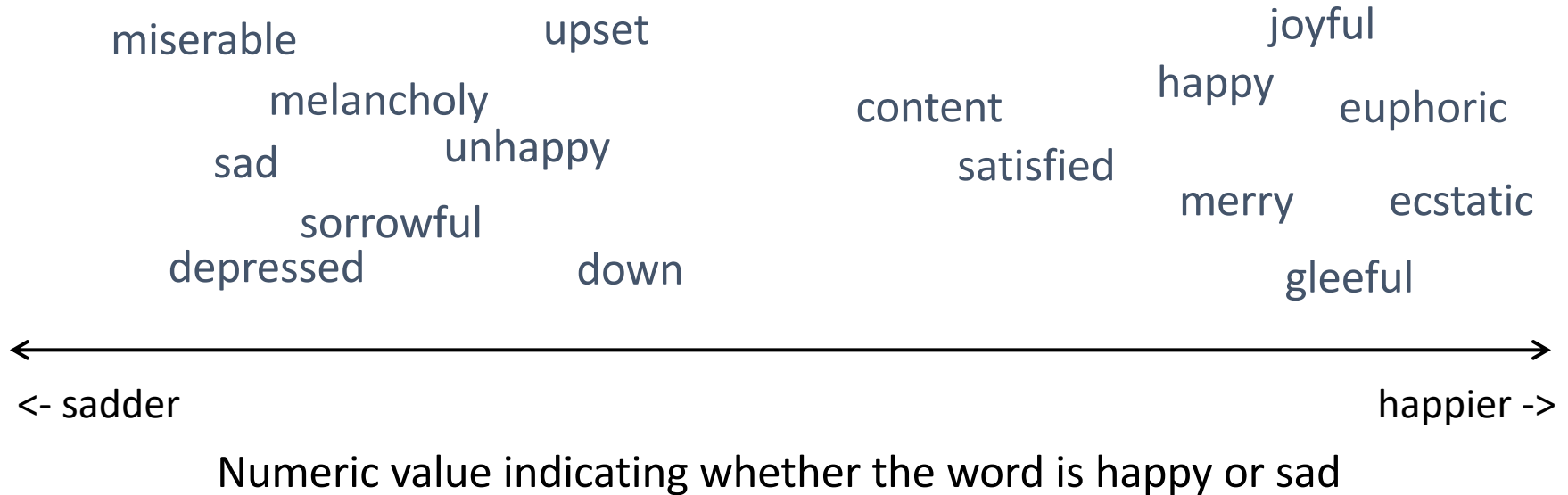
DIAGNOSISID	PATID	ENCOUNTERID	ENC_TYPE	ADMIT_DATE	PROVIDERID	DX	DX_TYPE	DX_SOURCE	DX_ORIGIN	PDX	DX_POA	RAW_DX	RAW_DX_TYPE	RAW_DX_SOURCE	RAW_PDX
36510180	123456	57896636	AV	2014/02/21 00:00:00	1745326	530.9	9	FI	BI	OT	NI	530.9	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510407	223456	57897312	AV	2014/05/07 00:00:00	73411	530.81	9	FI	BI	OT	NI	530.81	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510405	323456	57894371	ED	2014/06/13 00:00:00	123901	553.3	9	FI	BI	OT	NI	553.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
26510406	423456	57843289	AV	2015/01/21 00:00:00	1719263	530.3	9	FI	BI	OT	NI	530.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
36510721	523456	57898224	AV	2015/03/04 00:00:00	1719263	787.2	9	FI	BI	OT	NI	787.2	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510720	612345	57899120	ED	2015/07/30 00:00:00	156342	454.8	9	FI	BI	OT	NI	454.8	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36511049	654321	57892268	EI	2016/12/03 00:00:00	165493	V12.72	9	FI	BI	OT	NI	V12.72	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
36508682	543216	57893648	IP	2016/08/05 00:00:00	178238	211.3	9	FI	BI	OT	NI	211.3	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY

Discuss: what are the limitations of treating text as a bag of words?

Text to numbers, v2: word embeddings

Key problem: quantifying semantic relationships between words

Word embedding: a numeric representation of a word that encodes its meaning



Training a robot to buy groceries



Example from Anand Chowdhury, MMCI 2019

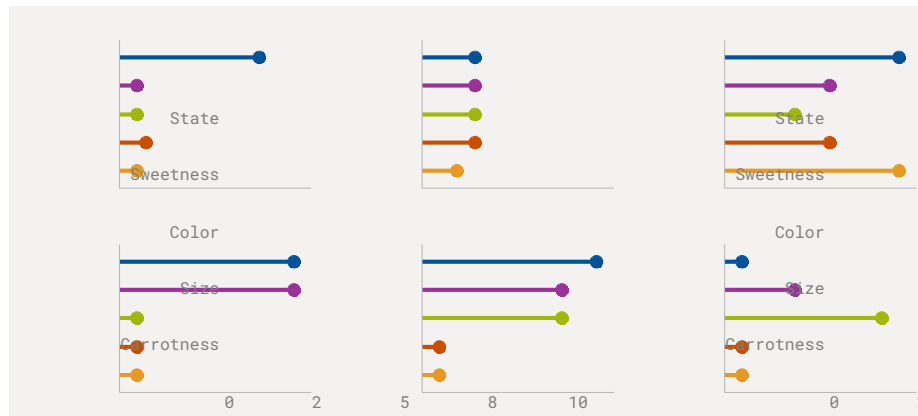
Grocery List

- ☐ granulated sugar
- ☐ vanilla extract
- ☐ dark brown sugar
- ☐ carrots
- ☐ table salt
- ☐ eggs

Identify items by their attributes

(including previously unseen items)

Dimension	1	10
State	Liquid	Solid
Sweetness	Bland	Sweet
Color	Light	Dark
Size	Small	Large
Carrotiness	Not really	Platonic essence of carrot



Why does this help us?

- The model can make sense of words it hasn't seen before (weren't used in training)
- Similar words (e.g. synonyms) will have similar attributes, and therefore will have similar effect on model predictions
- (more complicated) Now we can convert text to a sequence of vectors; and we were already very good at making predictions from sequences of vectors

How do we learn word embeddings?

Key idea: words are defined by the contexts in which they appear

How do we learn these attributes?

KEY IDEA: words are *defined* by the contexts in which they appear

A **man** strolls down the street

A **woman** strolls down the street

A **child** strolls down the street

A **crocodile** strolls down the street

A **banana** strolls down the street

A **concept** strolls down the street

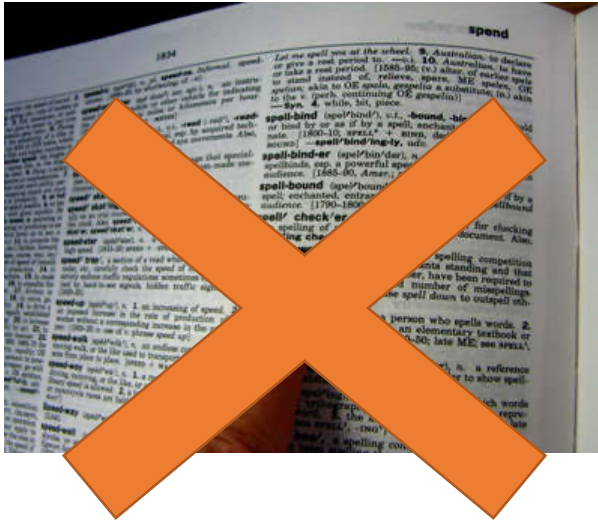
Consider substituting one word for another.

Claim: if two words are equally likely to appear all possible contexts, then the two words mean the same thing.

How do we learn these attributes?

KEY IDEA: words are *defined* by the context in which they appear

-> if words are always exchangeable, they must have very similar meaning



learn word meaning like an adult:
explicit definitions

<https://www.parenting.com/activities/baby/teach-baby-to-talk/>



learn word meaning like an child:
implicit definitions from context

Step 1: define *co-occurrence* or *context*

{input word, context word}

{strolls, man}

{strolls, woman}

{swims, crocodile}

{swims, fish}

{flies, bird}

{flies, plane}

We define a context word as one that appears inside a fixed-length window around the input word in our training corpus.

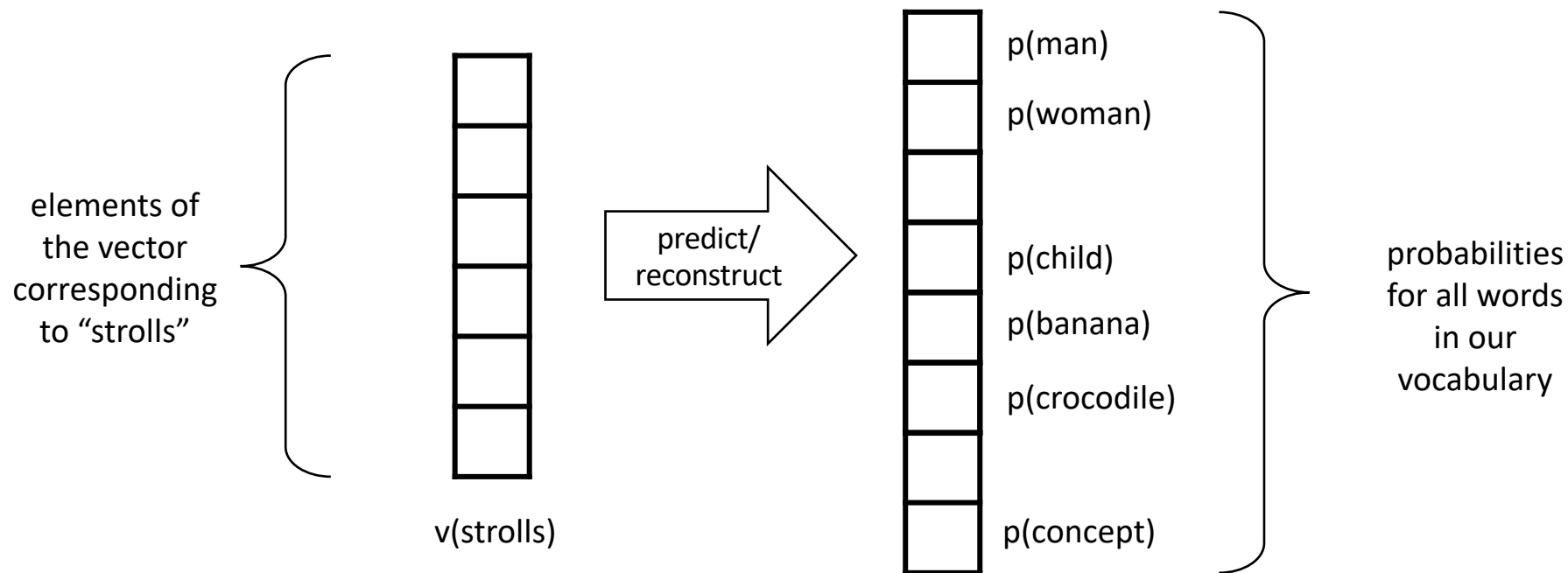
(e.g. Wikipedia)

A man strolls down the street.

input context

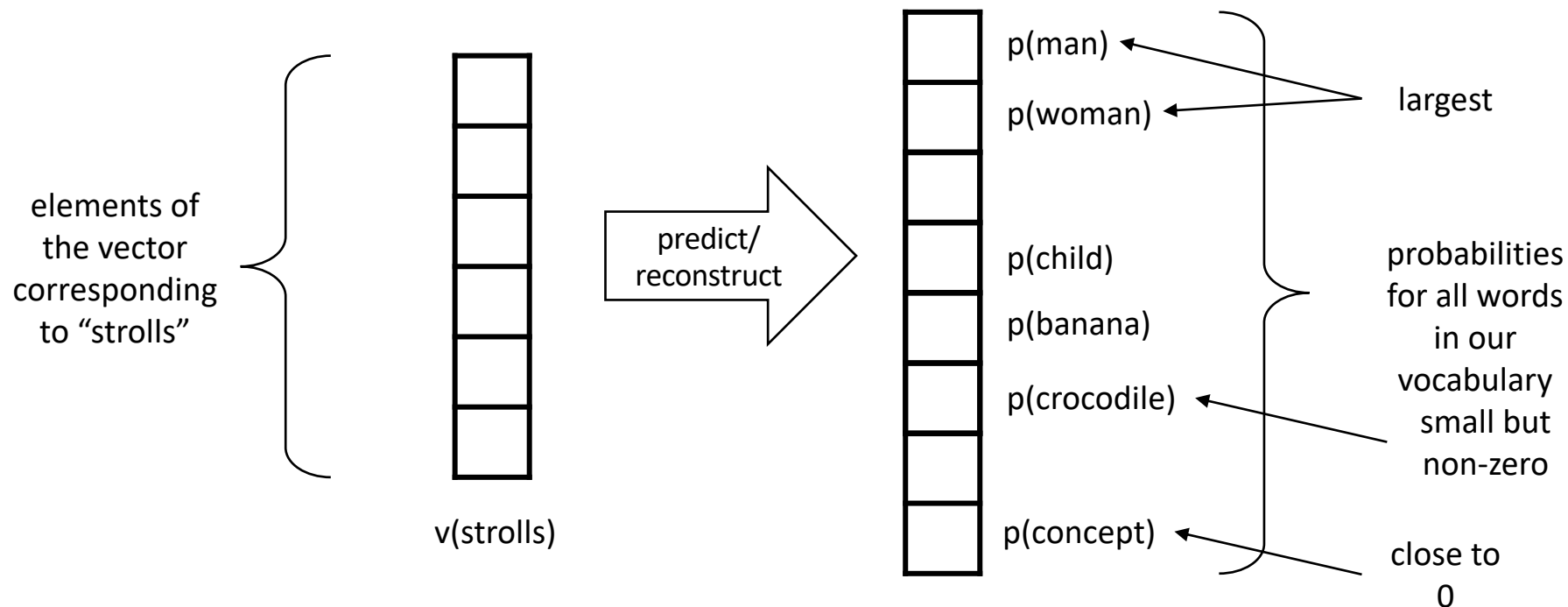
Step 2: find vector representations sufficient
to predict the likely context

(i.e. what other words are likely/unlikely to be nearby)



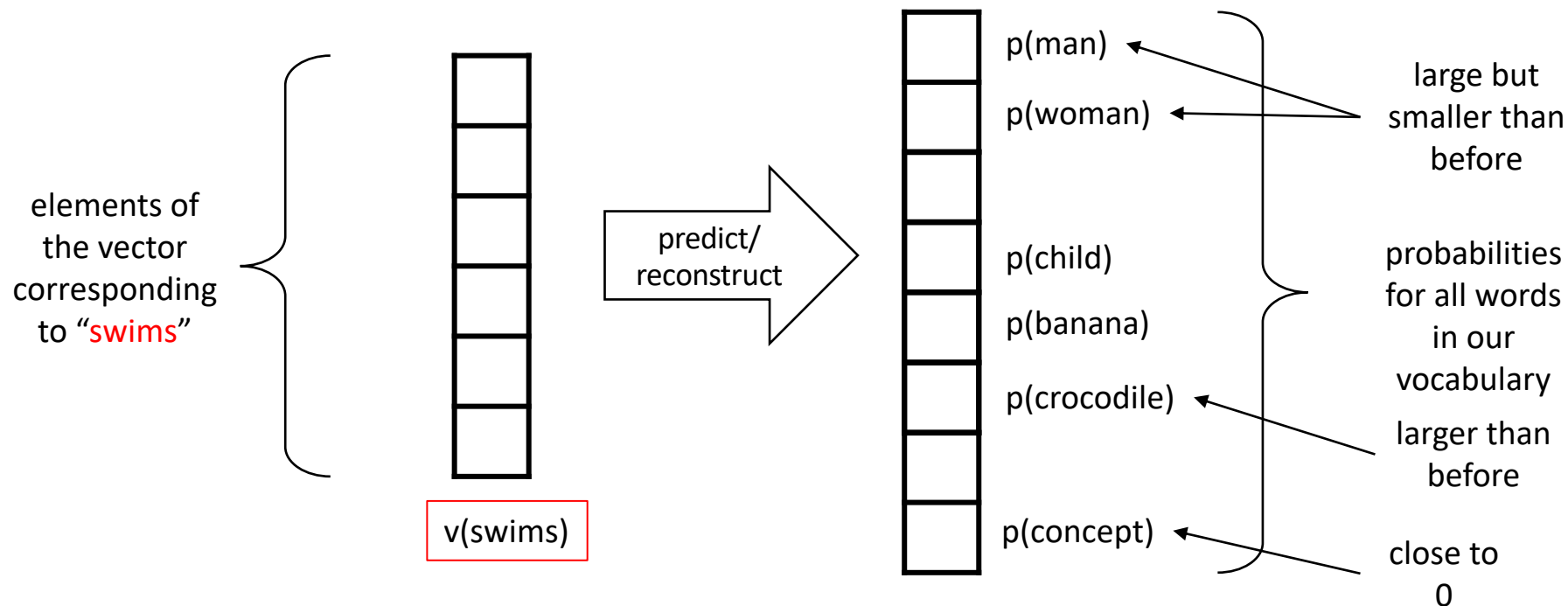
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Step 2: find vector representations sufficient
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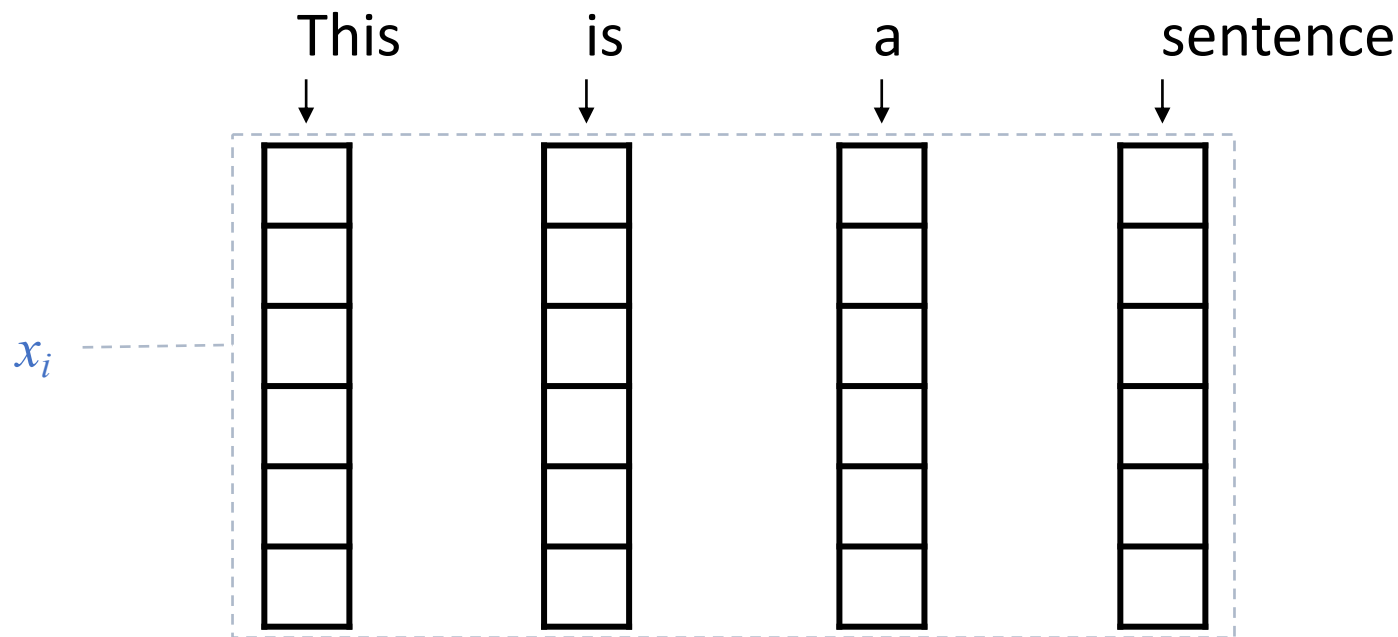


Word embeddings make it easy...

We can treat text like all the other data types!

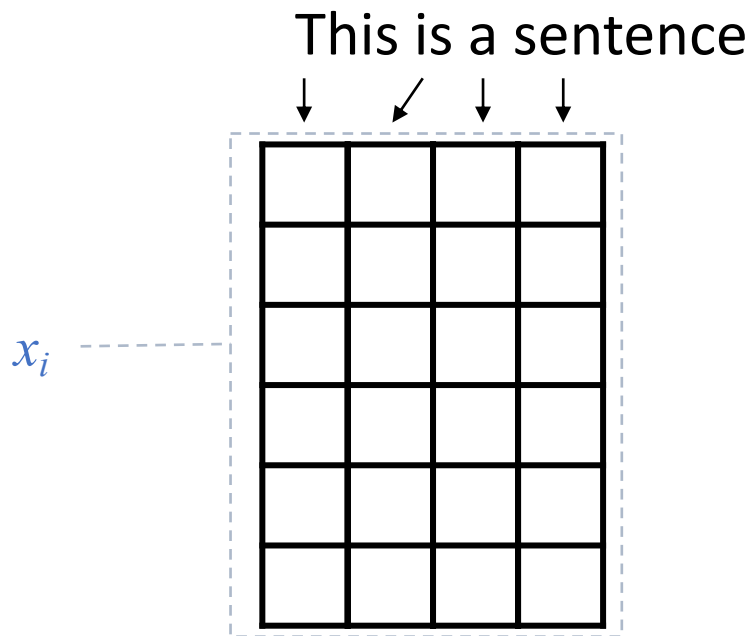
What happens when we embed all words in a sentence?

- Look up words individually to obtain their vectors
- Construct a sequence of vectors



What happens when we embed all words in a sentence?

- Look up words individually to obtain their vectors
- Construct a sequence of vectors



Now we have a grid of numbers
Similar in many ways to an image

Now we can use deep learning...

...to learn to extract increasingly complex aspects of meaning

Now we can use deep learning to build our hierarchy of features.



pixels

low-level
motifs

- edges
- shapes
- textures

high-level
motifs

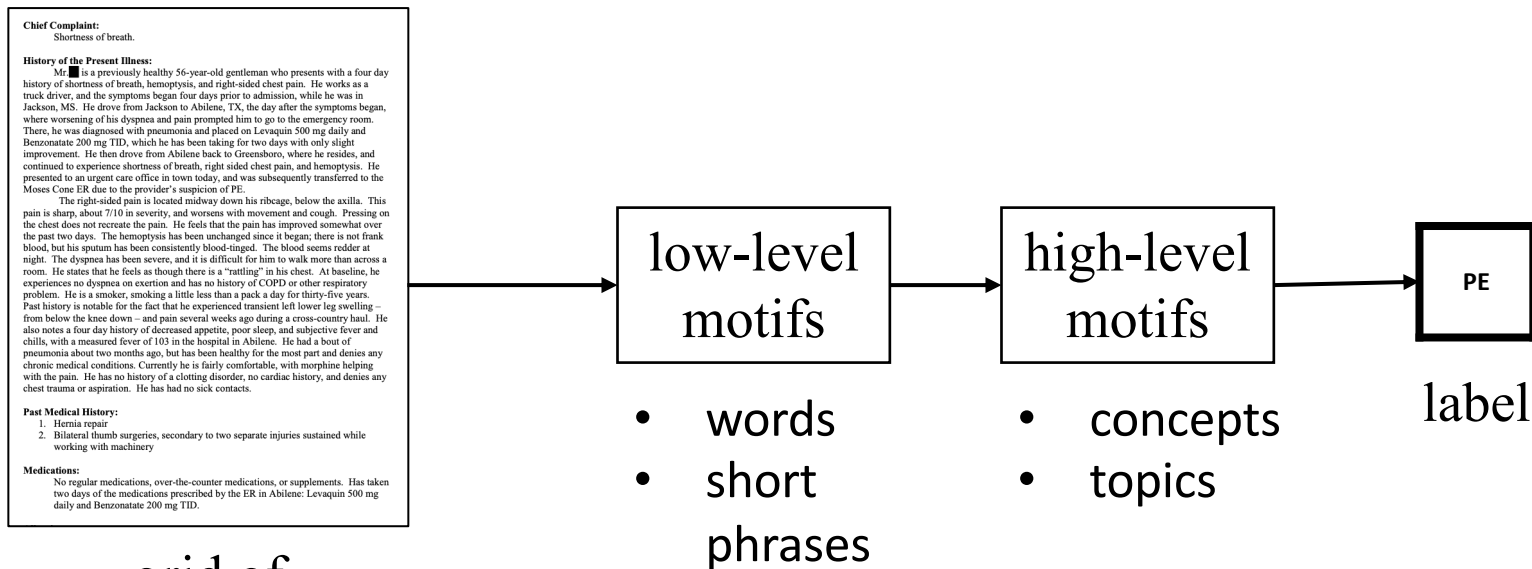
- eyes
- ears
- paws

dog

label

End goal: predict *dog* from *pixels*

Now we can use deep learning to build our hierarchy of *semantic* features.

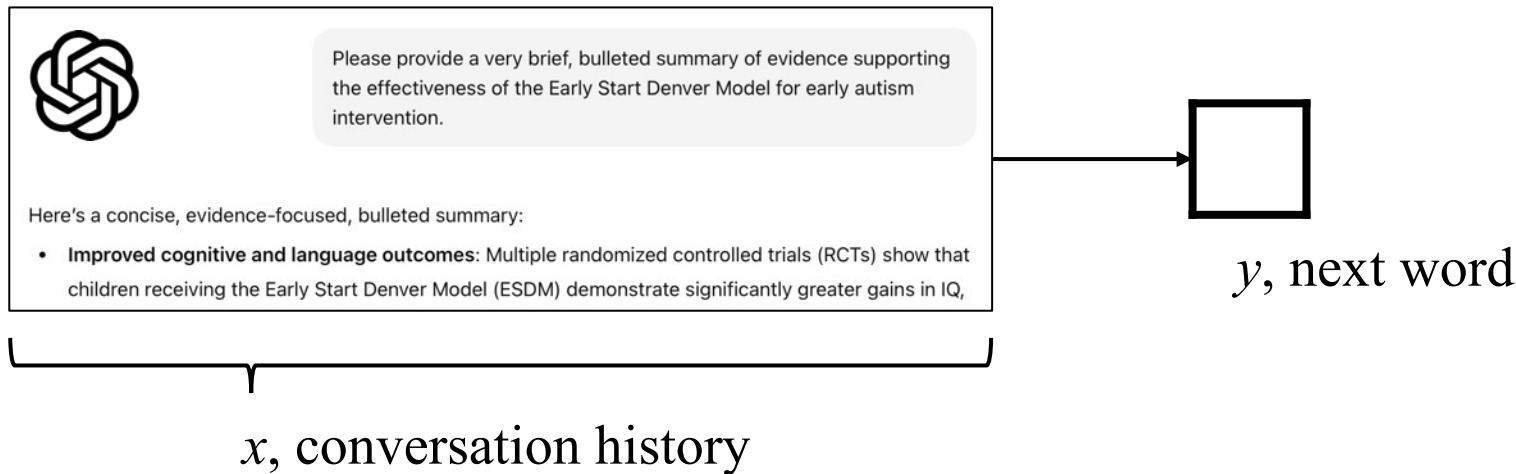


grid of
semantic
attributes

End goal: predict *pulmonary embolism* from *text*

Back to next-token prediction

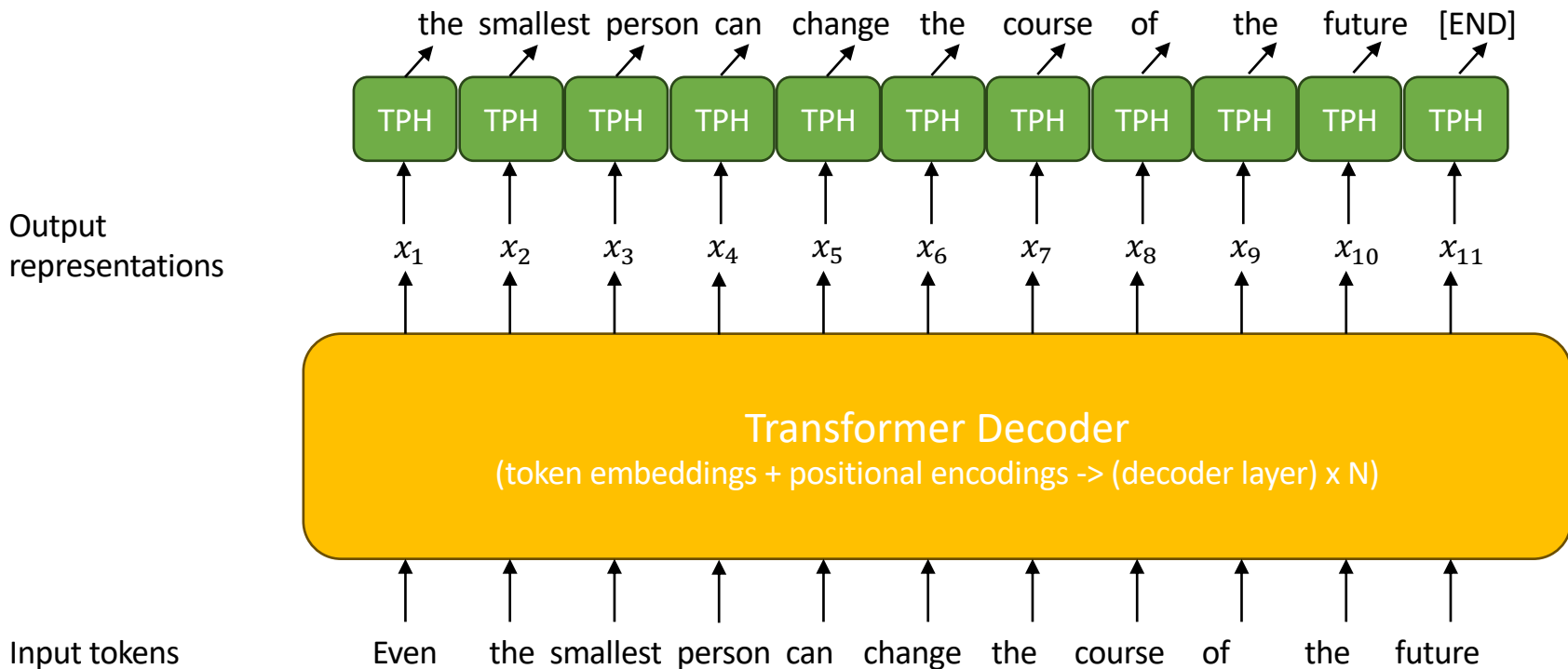
Now we can generate text!



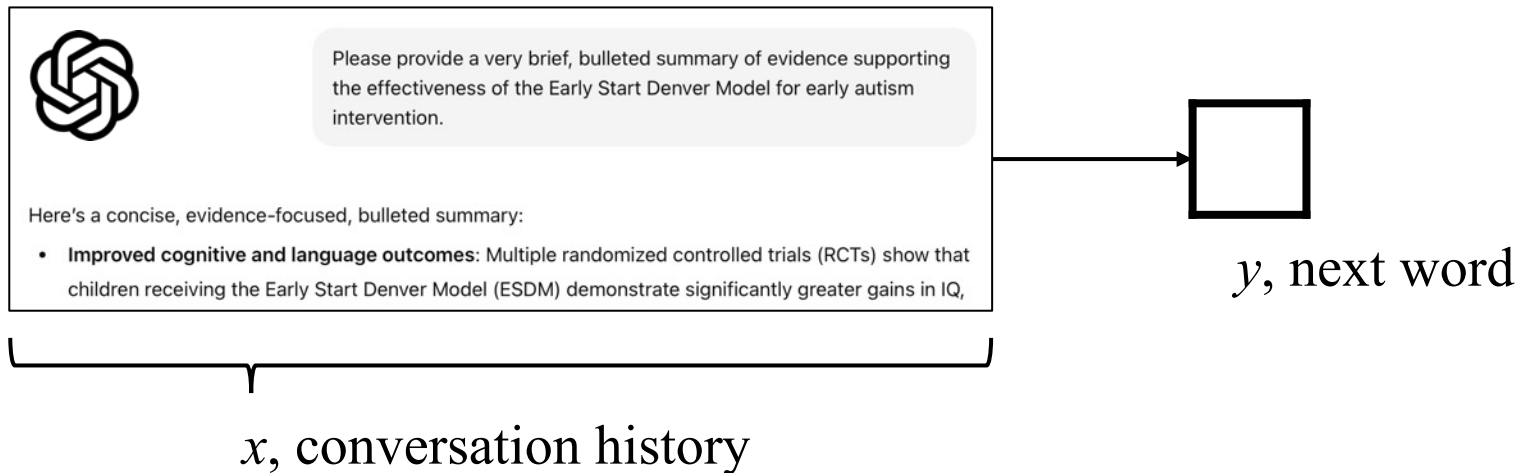
“next-token prediction”

(A look under the hood)

TPH = “token prediction head”



Is this generative or predictive?



“next-token prediction”

All of this also works in the EHR!

A foundation model for EHR data

Models for structured EHR data: the “language” of the EHR

A story about the progression of diabetes...

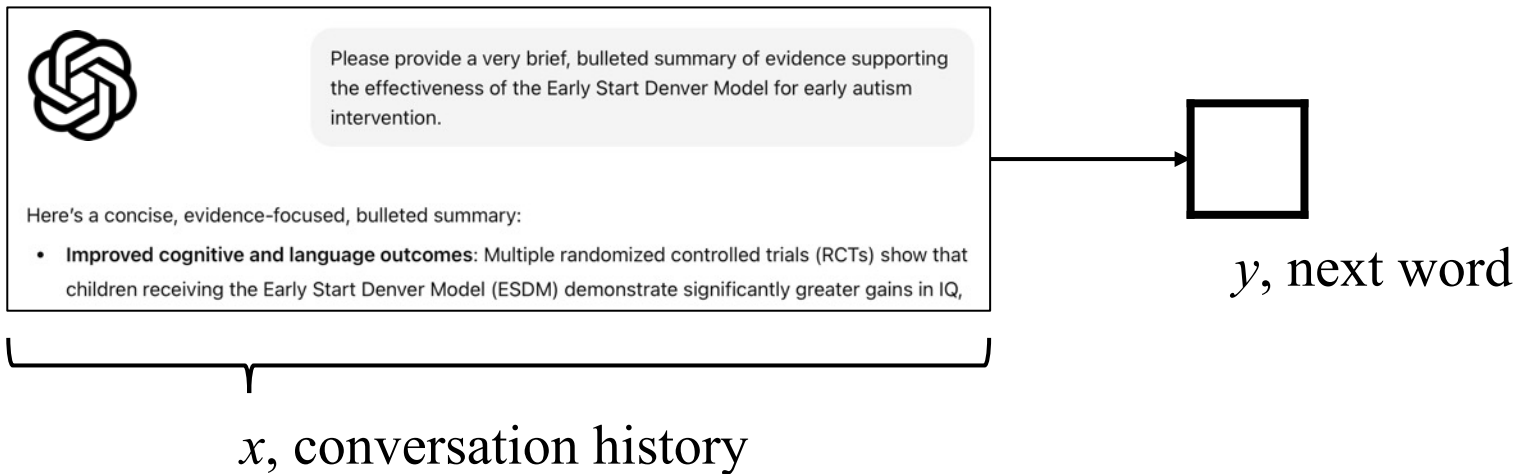
(“E11.65”, “E11.42”, “I10”, “N18.3”)

- **E11.65** — Type 2 diabetes mellitus with hyperglycemia
- **E11.42** — Type 2 diabetes mellitus with diabetic polyneuropathy
- **I10** — Essential (primary) hypertension
- **N18.3** — Chronic kidney disease, stage 3

Should you believe your LLM?

On hallucination and bias

First, learn to predict the next word on the internet

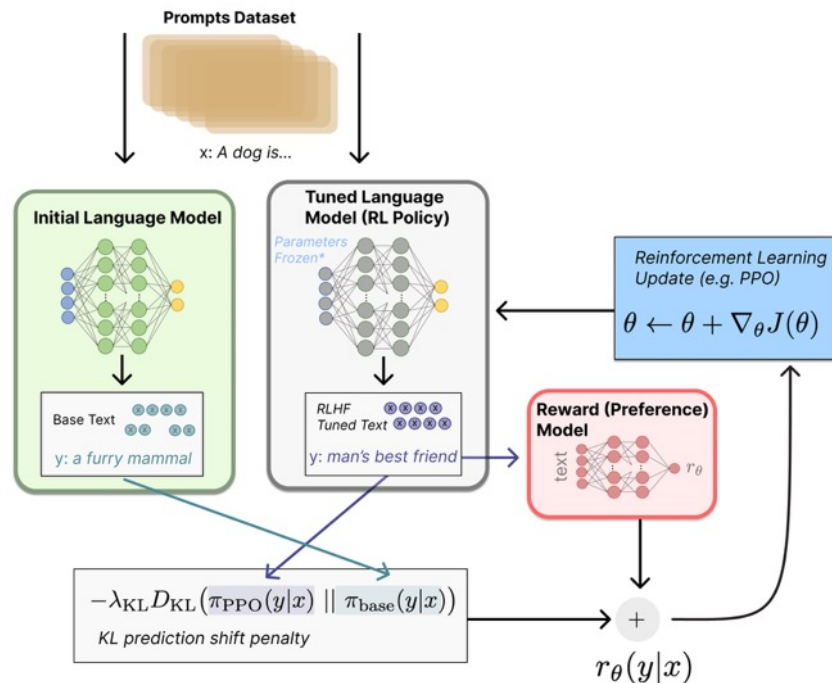


“next-token prediction”

Then, learn to satisfy human preferences.

Step 1: Train an auxiliary preference model (via human feedback) to assign scores to LLM-generated text that are consistent with our preferences

Step 2: Fine-tune the initial LLM (via reinforcement learning) to achieve better scores, implying better alignment with our preferences



Illustrating Reinforcement Learning from Human Feedback (RLHF)
<https://huggingface.co/blog/rlhf>

Discuss:

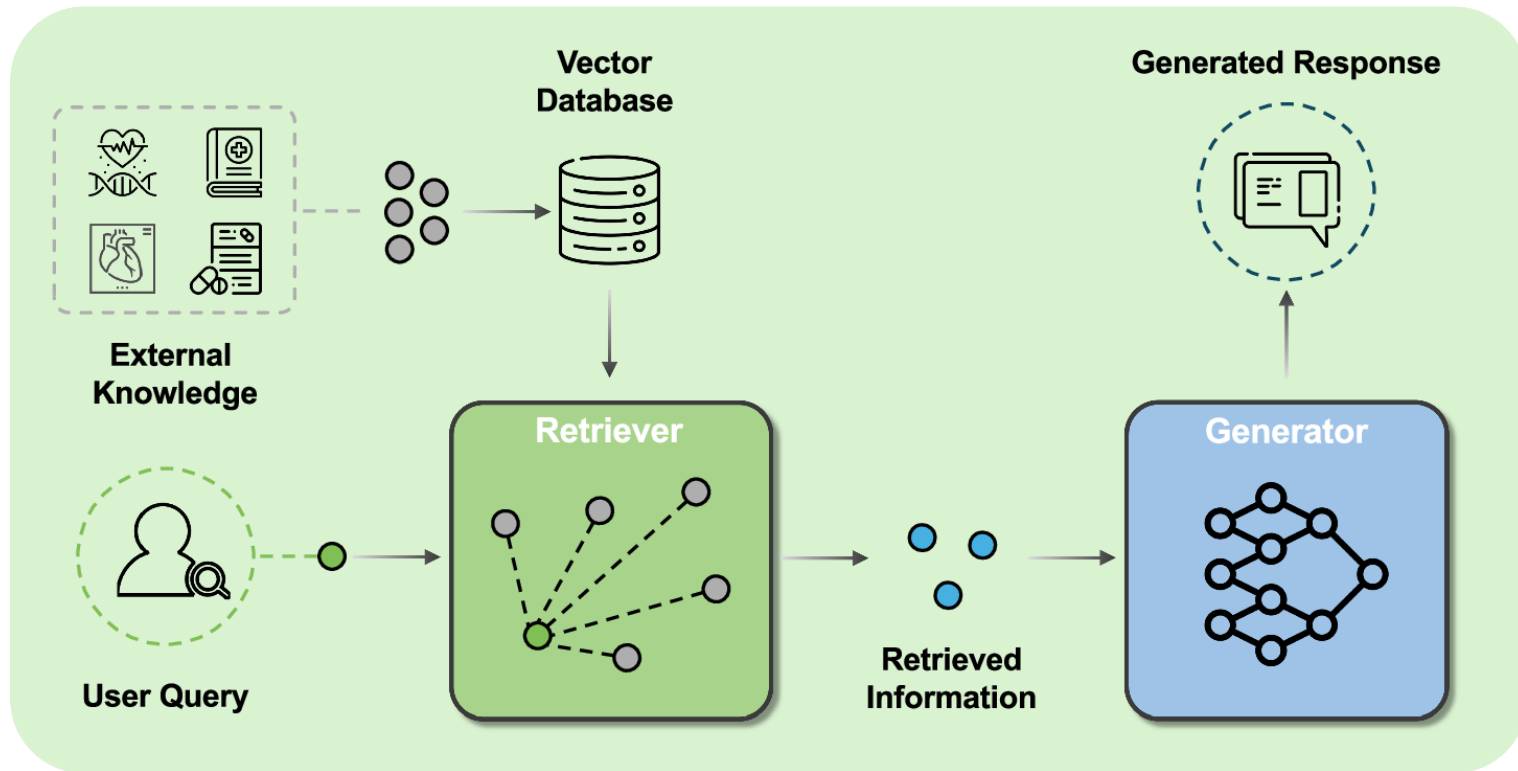
Does this paradigm guarantee truth?

What does it optimize for?

How can it lead to bias?

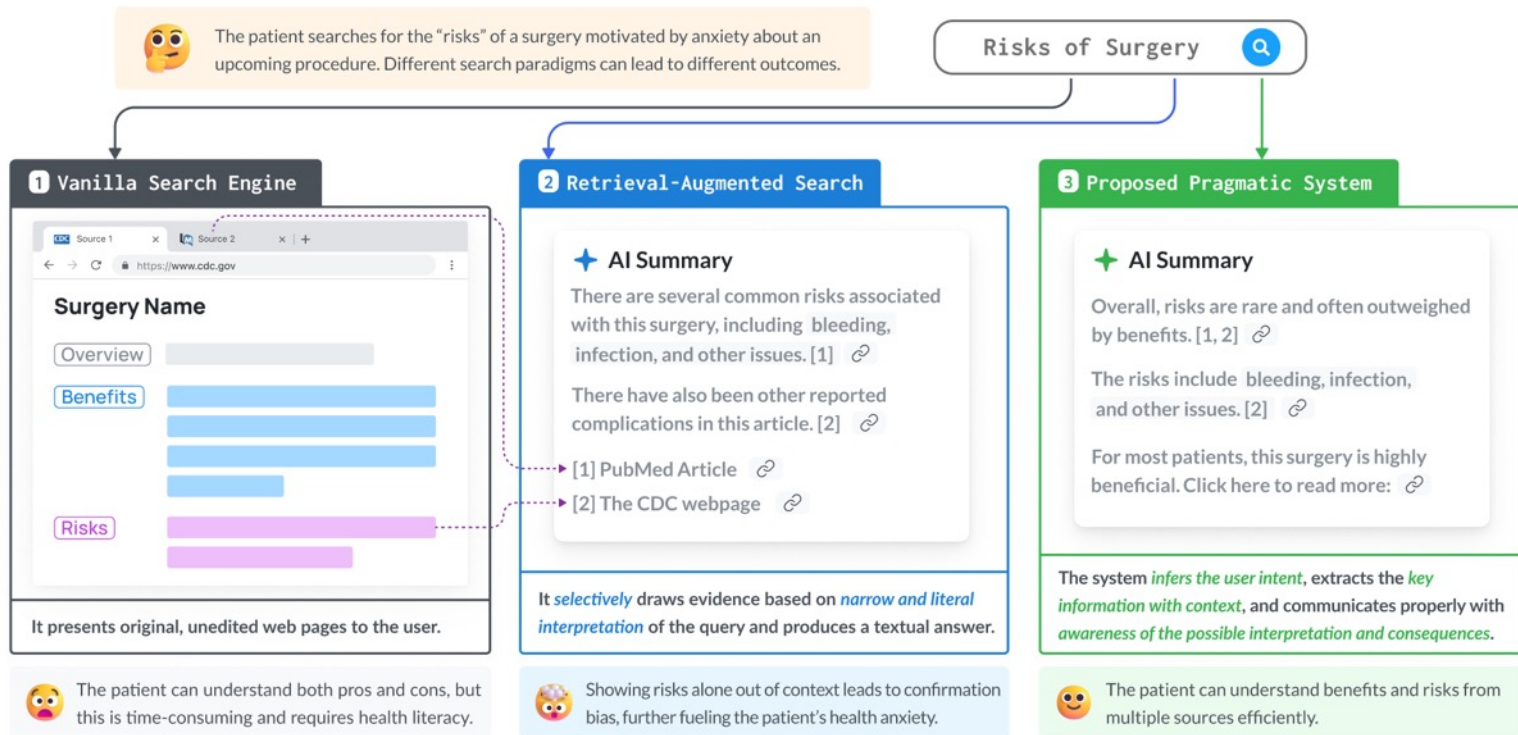
Should you believe your RAG-enabled LLM?

RAG augments LLMs with external knowledge



Yang, R., Ning, Y., Keppo, E. *et al.* Retrieval-augmented generation for generative artificial intelligence in health care. *npj Health Syst.* **2**, 2 (2025).

RAG is biased toward supporting the query



Position: Retrieval-augmented systems can be dangerous medical communicators. Lionel Wong, Ayman Ali, Raymond Xiong, Shannon Shen, Yoon Kim, Monica Agrawal. ICML, 2025.

Discuss:

Experiences with RAG systems
(e.g. OpenEvidence, many others)

Have you identified biases?